Mobile Validation Test Plan

Purpose: End-to-end mobile device validation of all 3 CompoundIQ applications before first real patient transaction.

Why this exists: All 6 rounds of Cowork QA ran in desktop browsers (1440px+ viewports). No mobile validation has been performed. Patient checkout is mobile-first by design (patients tap SMS links on phones), so this is the single highest-risk gap before production.

Time required: 30-45 minutes first pass. Re-run any time the `/checkout/*` routes, SessionBanner, or mobile-responsive components change.

Cost: $0 using your own devices. Optional: BrowserStack or LambdaTest (~$40/mo) if you don't have Android hardware.

Target URL

https://functional-medicine-infrastructure.vercel.app

Device Matrix

Validate on these device + browser combinations. **Priority 1** devices MUST pass before first real order; **Priority 2** are tracked as bugs but don't block.

Device	Browser	Priority	Coverage
iPhone (any model 12+)	Safari (default)	**P1**	Patient Checkout, Apple Pay
Android phone (any Samsung, Pixel, etc.)	Chrome (default)	**P1**	Patient Checkout, Google Pay
iPhone	Chrome iOS	P2	Checkout rendering sanity
iPad	Safari	P2	Clinic App graceful degradation

If you don't own one of the devices:

- **Missing Android?** Use BrowserStack live testing (real devices, ~$40/mo, cancel after testing) or borrow from someone
- **Missing iPhone?** Same — BrowserStack has real iPhones
- **No iPad?** Skip P2 iPad row; the clinic app is not designed for iPad and won't block launch

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Pre-Test Setup

Before you start testing:

1. Open the live app on your desktop and log in as `admin@sunrise-clinic.com` / the password in `POC_CLINIC_ADMIN_PASSWORD` (set in your local env)
2. Create a test prescription end-to-end:
    - Select Alex Demo + Sarah Chen → Continue

    - Pick any Favorite OR walk the Cascading Builder with Semaglutide → Strive Pharmacy
    - Click 2x multiplier → sign → Confirm & Send
3. Wait for redirect to `/dashboard?sent=1`
4. Click the most recent order → copy the checkout URL from the order detail drawer
    - (Or run `npx dotenv -e .env.local -- npx tsx scripts/get-checkout-url.ts` if you have the repo locally)
5. You will paste this URL into the mobile browser address bar manually (simulates the SMS tap)

Expected URL format: `https://functional-medicine-infrastructure.vercel.app/checkout/<order_uuid>?token=<jwt>`

Test Sections

PART A — Patient Checkout (iOS Safari) — PRIORITY 1

Open the checkout URL on your iPhone in Safari (default browser).

A1 — Page Load & Layout:

- [ ] Page loads without horizontal scroll (all content fits 320-428px viewport)
- [ ] Clinic name "Sunrise Functional Medicine" displays prominently at top
- [ ] Line item "Prescription Service" displays (NOT medication name)
- [ ] Total price matches what was set on desktop ($300.00 expected)
- [ ] Trust signals visible: "256-bit TLS Encryption", "Powered by Stripe"
- [ ] No medication name, dosage, or diagnosis appears anywhere on the page (HIPAA check)
- [ ] Footer reads "Your payment info is encrypted and never stored by CompoundIQ"

A2 — Form Fields:

- [ ] Email field renders with iOS keyboard showing @ key
- [ ] Tapping the email field does NOT cause the page to zoom in and get stuck (common iOS bug when font-size < 16px)
- [ ] Tab/Next key progression through form fields works

A3 — Stripe Payment Element:

- [ ] Stripe Elements iframe loads (card number, expiry, CVC inputs visible)
- [ ] **Apple Pay button appears at the top of the payment element** (this is the critical check)
- [ ] Card input autocompletes from iCloud Keychain if you have a saved card
- [ ] Typing in card number field works smoothly (no lag, no disappearing digits)

A4 — Apple Pay Flow:

- [ ] Tap Apple Pay button → Face ID / Touch ID prompt appears
- [ ] After biometric auth, payment sheet shows correct clinic name ("Sunrise Functional Medicine") and correct amount ($300.00)
- [ ] Sheet shows "Prescription Service" as the line item (not medication name)
- [ ] Cancel button works and returns to the checkout page without error

Do NOT complete the Apple Pay transaction during testing unless you want a real $300 charge to your card — the app is in Stripe test mode ONLY if the test Stripe publishable key is active. Verify this with desktop dev tools before tapping Pay.

If you ARE in test mode, proceed:

- [ ] Tap Pay → biometric confirmation → success animation in Apple Pay sheet → redirect to `/checkout/success/<order_id>`
- [ ] Success page: green checkmark, "Payment Received", amount in green, order reference (first 8 chars of UUID in monospace font)
- [ ] "What Happens Next" card renders with 3 steps
- [ ] No medication name on success page

A5 — Card Entry (fallback if skipping Apple Pay):

Use Stripe test card `4242 4242 4242 4242`, expiry `12/28`, CVC `123`, ZIP `78701`.

- [ ] Type in card number field — cursor doesn't jump around
- [ ] Form validation: invalid card number shows inline error, does not crash
- [ ] Submit button is tap-target friendly (min 44×44 px)
- [ ] Pay button submits and shows loading state
- [ ] Redirects to success page on completion

A6 — Expired Link:

Navigate to `https://functional-medicine-infrastructure.vercel.app/checkout/expired`

- [ ] Page renders without horizontal scroll
- [ ] Clock icon visible and properly sized
- [ ] "This payment link has expired" message displays
- [ ] Instruction to contact clinic
- [ ] Zero PHI (no order ID, no medication, no patient name)

PART B — Patient Checkout (Android Chrome) — PRIORITY 1

Repeat the full PART A test sequence on an Android phone in Chrome, with one substitution: **Google Pay instead of Apple Pay**.

Key differences to watch for:

- [ ] Google Pay button appears instead of Apple Pay at the top of the Stripe Elements payment area
- [ ] Tapping Google Pay opens the Google Pay sheet
- [ ] Sheet shows correct clinic name and amount
- [ ] Card entry with Stripe test card works the same as iOS
- [ ] Success page renders identically

Common Android-specific issues to watch for:

- Form fields zooming on focus (should not happen with properly set viewport)
- Chrome address bar hiding/showing causing layout jumps

- Back button behavior after success (should not re-submit payment)

PART C — Clinic App (iPad Safari) — PRIORITY 2

Open the live app on an iPad in Safari. Log in as `admin@sunrise-clinic.com` / the password in `POC_CLINIC_ADMIN_PASSWORD` (set in your local env).

C1 — Layout:

- [ ] Sidebar is visible or collapsible without breaking the main content
- [ ] Dashboard KPI cards are readable and not cramped
- [ ] Order table is horizontally scrollable if columns don't fit (should not be clipped)
- [ ] No overlapping text or buttons

C2 — Core Workflows:

Walk through the following. Each should complete without a hard crash or missing UI:

- [ ] New Prescription → Patient Search → Select Alex Demo → Continue
- [ ] Favorites tab shows at least one card; tapping a card navigates to the margin page
- [ ] Margin Builder: multiplier buttons are tap-friendly; sig field accepts input
- [ ] Save as Draft: button works, redirect to dashboard happens
- [ ] Drafts tab visible and clickable

C3 — Known Tolerable Issues (OK if they happen):

- Narrow viewport causes some text truncation
- Hover-only affordances (like pipeline row Actions menu) replaced by tap
- Virtual keyboard obscures form inputs — scroll into view should handle it

C4 — NOT tolerable:

- App refuses to load
- Login form unresponsive
- Cascading Builder dropdowns don't open on tap
- Any flow that blocks the save-as-draft path

PART D — Clinic App (iPhone Safari) — PRIORITY 2

Open the live app on iPhone in Safari. Log in as clinic admin.

Because the clinic app is NOT designed for phone, we're only checking for hard crashes and sign-of-life:

- [ ] Login form renders and accepts input
- [ ] Post-login redirect lands on /dashboard without error
- [ ] Dashboard shows SOMETHING useful (even if cramped)
- [ ] Attempting New Prescription does not crash the app

Log a P2 bug for anything broken here. Does NOT block launch.

PART E — Ops Dashboard (iPhone Safari) — PRIORITY 2

Open the live app on iPhone in Safari. Log in as `ops@compoundiq-poc.com` / the password in `POC_OPS_PASSWORD` (set in your local env).

- [ ] Login lands on /ops/pipeline without crash
- [ ] At least one order row is visible even if the table is horizontally scrollable
- [ ] Tabs (Pipeline, SLA, Adapters, Fax Queue, Catalog, Demo Tools) render
- [ ] /ops/demo-tools page: credential table is legible, Reset Credentials button is tappable (do NOT tap it during testing)

Same P2 rule: log bugs, don't block.

Results Template

Copy this into a new file when you run the test. Save as `mobile-validation-results-<DATE>.md` in `docs/qa-reports/`.

```
# Mobile Validation Results — <DATE>

**Tester:** <YOUR NAME>
**Devices used:**
- iPhone: <model>, iOS <version>
- Android: <model>, Android <version>, Chrome <version>
- iPad: <model>, iPadOS <version> (optional)

**Commit tested:** <git short sha at time of test>
**Live deployment:** https://functional-medicine-infrastructure.vercel.app

## Summary Table

| Section | Device | Priority | Result |
|---------|--------|----------|--------|
| A — Patient Checkout | iPhone Safari | P1 | PASS / FAIL / N/A |
| B — Patient Checkout | Android Chrome | P1 | PASS / FAIL / N/A |
| C — Clinic App | iPad Safari | P2 | PASS / DEGRADED / FAIL |
| D — Clinic App | iPhone Safari | P2 | PASS / DEGRADED / FAIL |
| E — Ops Dashboard | iPhone Safari | P2 | PASS / DEGRADED / FAIL |

## Issues Found

| ID | Priority | Device | Section | Description | Screenshot/Video |
|----|----------|--------|---------|-------------|------------------|
| M1 | | | | | |
| M2 | | | | | |

## Go / No-Go Recommendation

<PASS — ready for first real order OR FAIL — patient checkout blocker on <device> must be fixed>

## Notes
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```
<anything non-obvious, edge cases observed, recommendations>
```

Common Bugs to Watch For

From industry experience with mobile payment flows on Next.js + Stripe Elements:

1. **iOS Safari viewport zoom on input focus** — happens when font-size on an input is less than 16px. Fix: bump form input font-size to 16px minimum.
2. **Apple Pay button missing** — happens when the site is not served over HTTPS OR the Stripe key is for the wrong account. All our checkout URLs are on `functional-medicine-infrastructure.vercel.app` so this should be fine.
3. **Google Pay button missing** — Chrome requires a verified Payment Request manifest (Stripe handles this automatically for Stripe Checkout; for custom Stripe Elements integrations, the Payment Request button component needs to be correctly mounted).
4. **Form input keyboard covers submit button** — iOS Safari's keyboard floats over content. Fix: scroll-into-view on focus, or sticky submit button with `position: fixed; bottom: 0`.
5. **Success page does not render on redirect** — happens when the Next.js middleware redirects in a way that Safari blocks due to third-party cookies. Our middleware is same-origin so unlikely.
6. **Signature pad on iOS does not accept touch drags** — `react-signature-canvas` has had this issue; fix is `pointerEvents: 'auto'` explicitly. We don't need this for checkout (signing is desktop-only) but worth knowing for future clinic-app-mobile work.

What To Do If You Find a P1 Failure

1. Screenshot / video the failure on the device
2. Note the EXACT reproduction steps
3. Log as a new work order: `WO-MOB-<n>: <short description>`
4. Block the first real order until the WO is closed
5. Add a regression test (even a manual one in this doc) for the next mobile validation pass

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Automation Future State (NOT for first pass)

Post-launch (60+ days), consider automating this with:

- **Playwright mobile emulation** in CI — covers viewport/layout, but NOT Apple Pay / Google Pay
- **BrowserStack Automate** — real devices in CI, ~$150/mo for cloud devices
- **Percy or Chromatic** — visual regression on mobile viewports

For now: manual testing every time the checkout flow changes. It's 30 minutes. It will catch 95% of what automation would.

Version History

- v1.0 — 2026-04-19 — Initial plan created after mobile gap flagged during pre-launch review